



21 DAYS SQL CHALLENGE

DAY 9

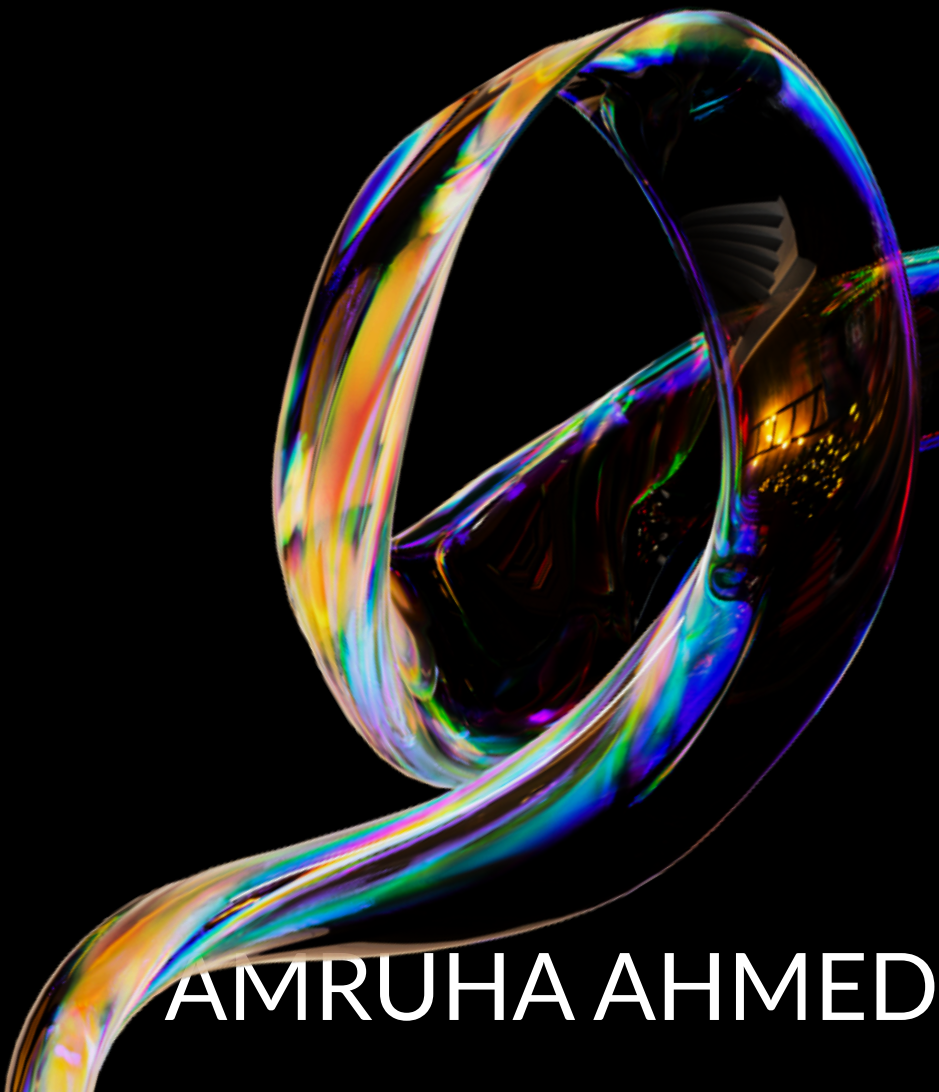
BY AMRUHA AHMED

DAILY CHALLENGE

Calculate the average length of stay (in days) for each service, showing only services where the average stay is more than 7 days. Also show the count of patients and order by average stay descending.

```
SELECT
service,
ROUND(AVG(DATEDIFF(departure_date,arrival_date))) AS Average_Length_Of_Stay,
COUNT(patient_id) AS "Number of Patients"
FROM patients
GROUP BY service
HAVING Average_Length_Of_Stay>7
ORDER BY Average_Length_Of_Stay DESC;
```

service	Average_Length_Of_Stay	Number of Patients
surgery	8	254
ICU	8	241



PRACTICE QUESTION 1 METHOD 1

Extract the year from all patient arrival dates.

```
SELECT YEAR(arrival_date) AS "Arrival Year"  
FROM patients;
```

Arrival Year
2025
2025
2025
2025



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PRACTICE QUESTION 1 METHOD 2

Extract the year from all patient arrival dates.

```
SELECT EXTRACT(YEAR FROM arrival_date) AS "Arrival Year"  
FROM patients;
```

Arrival Year
2025
2025
2025
2025



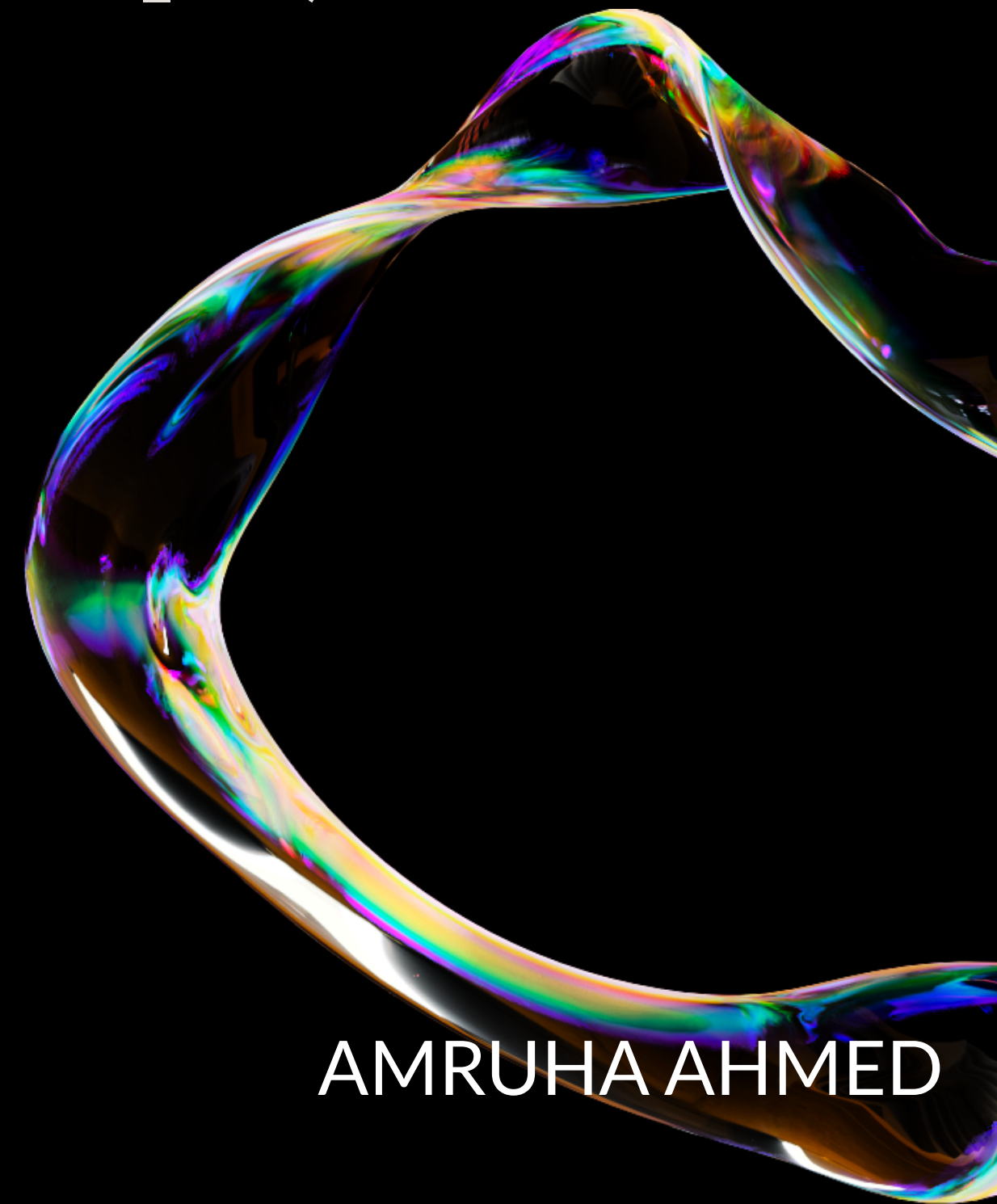
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PRACTICE QUESTION 2

Calculate the length of stay for each patient (departure_date - arrival_date)

```
SELECT  
patient_id,  
DATEDIFF(departure_date,arrival_date) AS "Length of Stay"  
FROM patients;
```

patient_id	Length of Stay
PAT-09484753	6
PAT-f0644084	1
PAT-ac6162e4	6
PAT-3dda2bb5	11



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PRACTICE QUESTION 3 METHOD 1

Find all patients who arrived in a specific month.

```
SELECT  
  patient_id,  
  arrival_date,  
  MONTH(arrival_date) AS "Month"  
FROM patients  
WHERE MONTH(ARRIVAL_DATE)=1;
```

patient_id	arrival_date	Month
PAT-003ce690	2025-01-19	1
PAT-8029775b	2025-01-29	1
PAT-4275a12b	2025-01-25	1

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PRACTICE QUESTION 3 METHOD 2

Find all patients who arrived in a specific month.

```
SELECT  
  patient_id,  
  arrival_date,  
  EXTRACT(MONTH FROM arrival_date) AS "month"  
FROM patients  
WHERE EXTRACT(MONTH FROM arrival_date)=1;
```

patient_id	arrival_date	month
PAT-003ce690	2025-01-19	1
PAT-8029775b	2025-01-29	1
PAT-4275a12b	2025-01-25	1

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The background is black with two abstract, colorful liquid-like swirls. One swirl is in the top right corner, and the other is in the bottom left corner. They have a rainbow-like iridescent sheen.

THANK YOU

for your time and attention

AMRUHA AHMED